CSS 430: Operating Systems - Syllabus

Instructor: Dr. Yusuf Pisan, pisan@uw.edu UW1-260Q

Class: TTh 11-1pm. UW2-131

Office Hours: Monday 2:15-3:15pm, Thursday 1:15-2:15pm at UW1-260Q. Email me 24 hours ahead of time to guarantee that I will be in my office, otherwise I might be grabbing coffee or wondering around campus.

Class Discord: [invite link]

Course Description

Principles of operating systems, including process management, memory management, auxiliary storage management, and resource allocation. Focus on the structure of the popular desktop and real-time operating systems. May not be repeated. Prerequisite: a minimum grade of 2.0 in CSS 343.

Learning Objectives

This course contributes to the following ABET outcomes:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, , and economic factors

3. an ability to communicate effectively with a range of audiences

4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Goals

As with most technical courses, besides ability and motivation, it takes time to learn and master the subject. No one succeeds without practice! Expect to spend an average of 15 hours a week outside of class time for this course; some of you may spend more time, some less time.

Textbooks

Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau https://pages.cs.wisc.edu/~remzi/OSTEP/

Grading

- In-Class Exercises: 10%

- Quizzes: 10%

- Homeworks: 10%

- Projects: 30%

- Exam-1: 20%

- Exam-2: 20%

There is no final exam. Exams are not cumulative.

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Assignment Submission: In-class Exercises will be graded as Complete/Incomplete and completed during the class hour. You can miss 2 in-class exercises without any penalty. You do not have to tell me when or why you will mis them. I will assume you have a good reason. If you miss more than 2, your in-class exercise grade will go down.

For projects, you can submit one and only one project 24 hours late without any penalty. If you are using your 24 hour extension, post a brief explanation to "Using Extension" assignment.

Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK.

Using AI to review topics, understand concepts is OK; giving homework questions to AI to answer is not OK.

It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct:
http://www.washington.edu/cssc/for-students/student-code-of-conduct/

Assignments must be submitted on-time. I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Access and Accommodations: Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact DRS at 425-352-5307 or uwbdrs@uw.edu. DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s), and DRS. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law.

Religious Accommodation Policy:
See https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/

Problems: If you are having difficulties, come and talk to me. If I don’t know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

See the School of STEM Course Policies, which covers:

- Academic Integrity
- Access and Accommodations

- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete
- Inclement Weather
- Parenting Resources
- Religious Accommodations
- Respect for Diversity
- Student Support Services
- Surviving Sexual and Relationship Violence
- Wonder How to Address Faculty?

Course Calendar

Week	Topics	Notes
Week-1 6 Jan 8 Jan	Introduction & Process Ch 1-6	Chapter quizzes due on Sunday each week.
Week-2 13 Jan 15 Jan	Scheduling Ch 7-10	HW1
Week-3 20 Jan 22 Jan	Memory Ch 12-15	Project-1
Week-4 27 Jan 29 Jan	Segmentation / Free Space Ch 16-17 Paging Ch 18-20	HW2
Week-5 3 Feb 5 Feb	Beyond Physical Memory Ch 21-23 Review	Project-2
Week-6 10 Feb 12 Feb	**Exam-1** Concurrency Ch 25-27	HW3
Week-7 17 Feb 19 Feb	Locks / Semaphores Ch 28-31	Project-3
Week-8 24 Feb 26 Feb	Common Concurrency Problems Ch 32	HW4
Week-9 3 Mar 5 Mar	Persistence Ch 35-38 File Systems Ch 39-42	Project-4
Week-10 10 Mar 12 Mar	Review **Exam-2**	
Finals week	No final exam (ignore what MyUW says)	

Operating Systems: Three Easy Pieces

https://pages.cs.wisc.edu/~remzi/OSTEP/

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